Digital Machines, Intelligence & Consciousness

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I propose to consider the question, “Can machines think?”

This should begin with definitions of the meaning of the terms “machine” and “think”.

Alan Turing in 1950

Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, 59(236), 433–460.

Turing replaced the question “Can machines think?” with a behavioural one:

Can a digital machine behave intelligently enough to be indistinguishable from a human in conversation?

Digital Computer Primer

(Binary) Turing Machine Tape

Read/Write head
(Head can shift Left or Right or not at all)

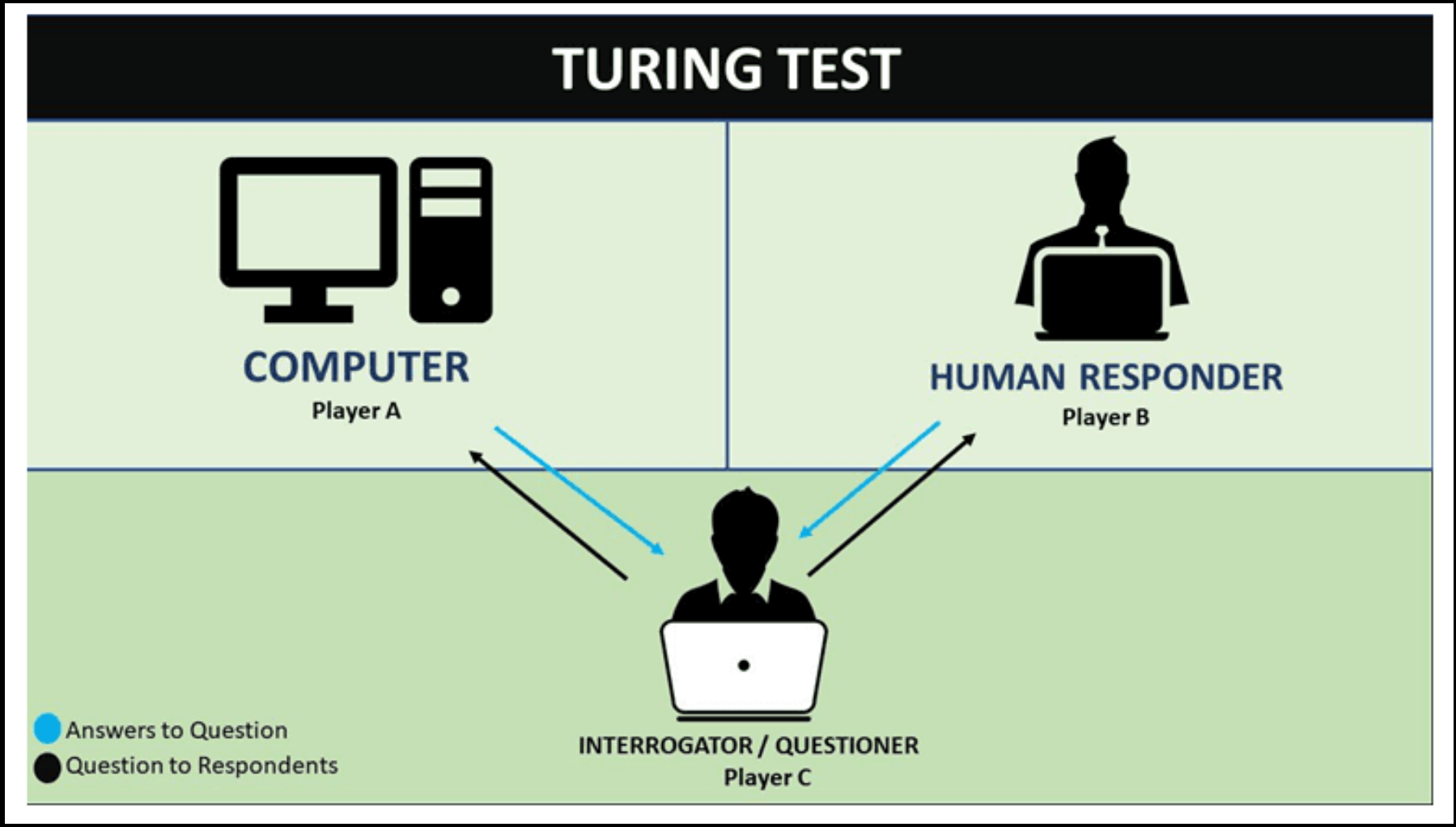
PROGRAMMING CODE in CARDS: C0, C1, C2,C3…

C1

Input character	Character to write	Move head (0=left, 1=right)	Next card
0	1	0	2
1	1	1	3

- C1 is the start card
- C0 is stop card

Imitation game (παιχνίδι μίμησης)



https://www.youtube.com/watch?v=4VROUIAF2Do

Turing argues that **digital computers**—machines capable of following programmable rules to manipulate symbols—are the right class of machines to consider for this question.

He claims that it is conceivable such computers could be made to perform so well in the imitation game that people would judge them **intelligent, even if they are simply executing instructions.**

Addressing Objections

Turing responded to a range of objections to machine intelligence—philosophical, theological, and logical—such as the claim that machines can only do what they are programmed to do, or that thinking requires a soul.

While he didn't fully "prove" machines can think, he argued these objections fail to rule out the possibility of intelligent machines.

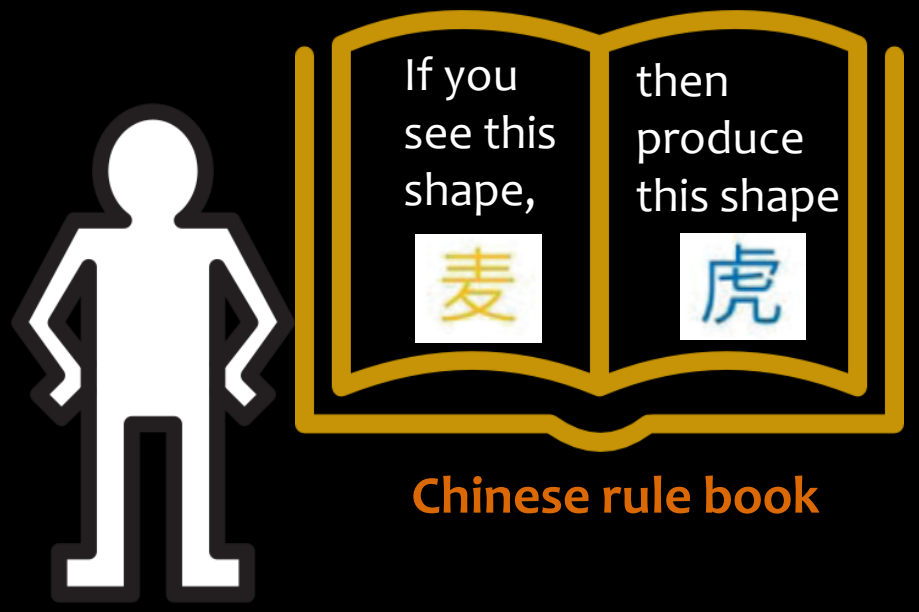
Learning machines

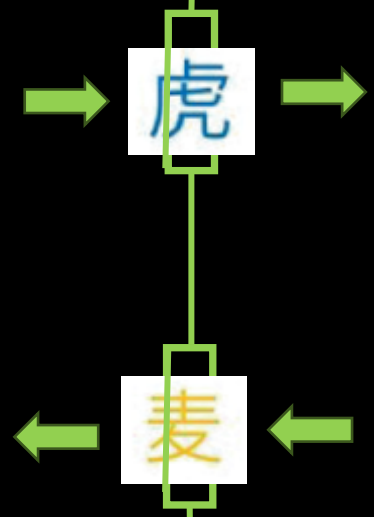
Turing introduced the idea that machines could learn—similar to how human minds develop—by starting with simple initial states and acquiring knowledge over time.

Chinese Room Test

Defined by John Searle as a counter-argument to the Turing test.

A person who **knows only English** is locked in a room.





Outside the room, a **native Chinese** speaker slip **questions written in Chinese** through a slot.

From the outside, it appears as if the room "understands" Chinese—but internal to the room, there is **no understanding**, only **symbol manipulation**.

https://www.youtube.com/watch?v=TryOC83PH1g

Syntax Vs Semantics

Syntax refers to the **structure and rules** that govern how symbols, words, or phrases are *arranged* in a language (whether natural or artificial). It's about *form* and *correct structure*, not meaning.

Semantics is about meaning — what sentences, phrases, or symbols mean or convey. It's not enough for a sentence to be structured correctly; it must make sense to be semantically meaningful.

Searle uses this philosophical thought experiment to argue:

- A **digital computer** manipulates symbols (syntax) based on rules, but it doesn't *know* what those symbols mean (semantics).
- So even if a computer behaves as if it understands language (for example, passes the Turing test), it doesn't mean it actually understands anything in the human sense.
- Therefore, running the right program **is not sufficient** for genuine understanding or consciousness.

Penrose's "Non-Computability" argument

If cognition (especially understanding) is non-algorithmic (non-computable), then consciousness itself cannot be simulated by a Turing machine.

The argument is primarily based on Kurt Gödel's incompleteness theorems and leads to the conclusion that a new, non-computable form of physics, specifically involving quantum mechanics in the brain, is necessary to explain the mind.

The hard problem of consciousness

David John Chalmers distinguishes:

- **Easy problems:** behavior, information processing, attention, learning
- **Hard problem:** why any of that is accompanied by **subjective experience**

Even if we had a *perfect algorithmic model* of cognition, the hard problem would still remain (the computation alone doesn't explain why there is experience).

Dennett's rejection of the "hard problem"

Daniel Dennett argues:

- **There is no "hard problem"** – only the illusion of one, created by Cartesian intuitions
- **Consciousness is not a special, non-physical substance** – it's an emergent functional property of complex information processing
- Once we abandon the intuition that consciousness is something *over and above* physical/computational processes, the "hard problem" dissolves. There's no mystery about "why processing feels like something" – that's a confused question based on false assumptions.

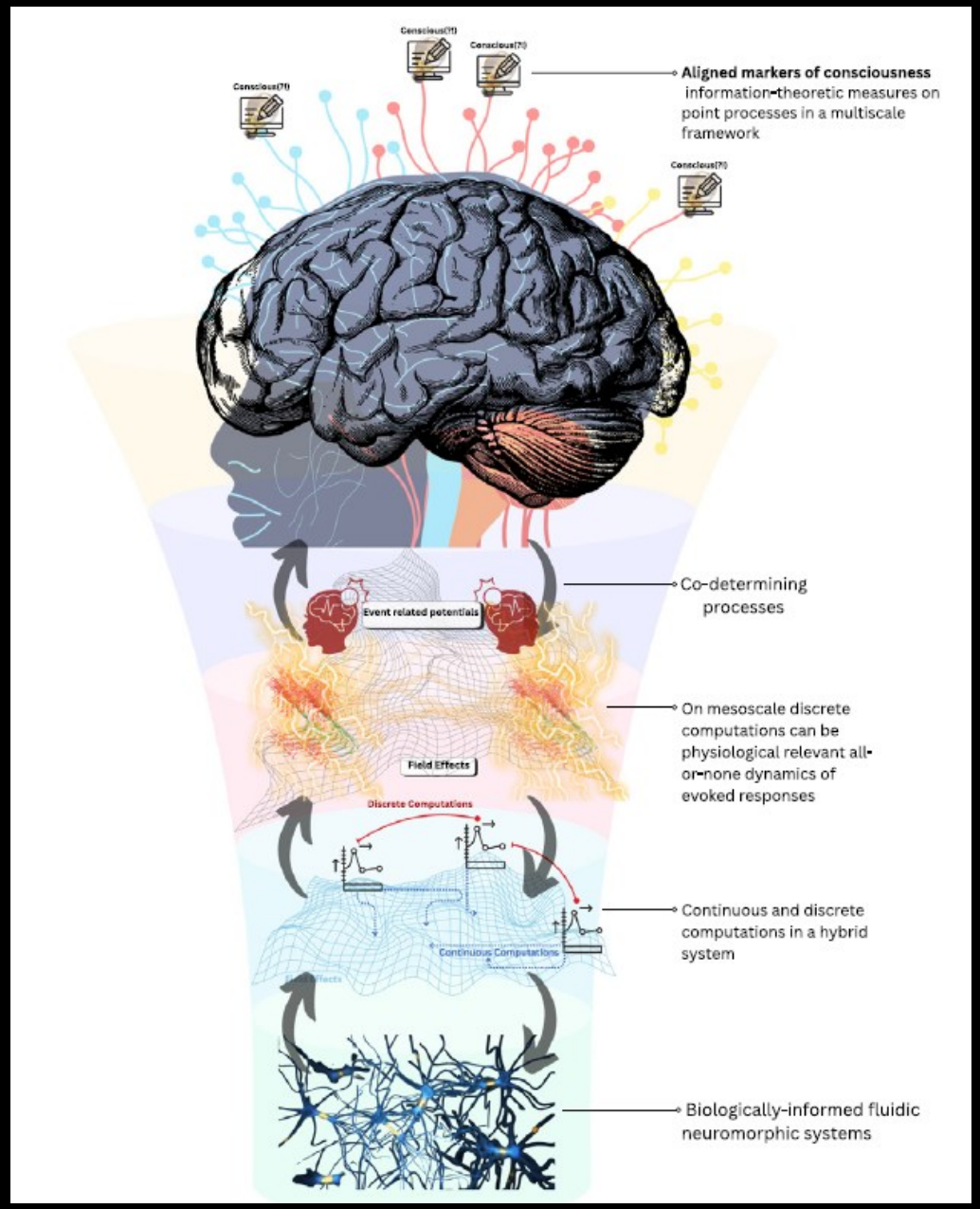
Biological computationalism

Borjan Milinkovic and Jaan Aru argue that:

The absence of consciousness in artificial systems is not merely due to missing functional organisation but reflects a deeper divide between digital and biological modes of computation and the dynamico-structural dependencies of living organisms.

https://doi.org/10.1016/j.neubiorev.2025.106524

https://www.youtube.com/watch?v=qWr__AY4Ask



What is the crucial difference between intelligence and consciousness?

Intelligence is the ability to **solve problems** like playing chess, diagnosing diseases, or driving a car, whereas consciousness is the ability to **experience emotions** such as pain, pleasure, love, and happiness.
Understanding this distinction is essential, especially in the context of AI, which currently exhibits intelligence but lacks consciousness.

Thinker	Core Concept	Main Argument	Can AI reach “ASI"?	Can ASI be "Conscious"?
Alan Turing	**The Turing Test**	If a system functions as if it is intelligent, then it possesses intelligence (Turing Test).	**Yes (pragmatically).** If it passes the Turing test, it's intelligent	**Agnostic.** The test addresses behavior, not inner subjective experience. Turing sidesteps the consciousness question.
John Searle	**The Chinese Room**	Symbol manipulation (syntax) alone cannot produce understanding (semantics) . A system can follow rules perfectly without comprehending meaning. Syntax ≠ semantics.	**Yes (functionally).** AI can exhibit intelligent behavior and solve problems as simulation.	**No (for Digital AI).** Consciousness is a genuine biological phenomenon. AI lacks understanding, and subjective experience. A simulation of a process is not the process itself. You cannot get "wet" from a computer simulation of water.
Roger Penrose	**Gödelian Argument & Orch-OR Theory**	Human consciousness involves non-computable processes (demonstrated via Gödel's incompleteness theorems), potentially grounded in quantum effects in neuronal microtubules. Digital computers are limited to algorithmic computation.	**No (full).** AI can be highly intelligent in specific algorithmic tasks, but it would always lack the 'non-computable' aspects of human thought.	**No.** True consciousness requires non-computable processes, possibly grounded in quantum physics, beyond current computational AI.
David Chalmers	**Property Dualism/ The Hard Problem**	There is a gap between processing data (Easy Problems—function, behavior, cognition) and subjective experience (Hard Problem—phenomenal consciousness, qualia, "what it's like" to experience).	**Yes.** Functional intelligence is achievable computationally. The "easy problems" can be solved algorithmically.	**Maybe.** Phenomenal consciousness (subjective experience, "what it's like" to be the AI) remains the hard problem—still open.
Philip Goff	**Panpsychism**	Consciousness (or proto-mental properties) is a fundamental quality of all matter, not just complex brains.	**Yes.** Matter can be organized into super-structures.	**Unclear.** All matter has proto-experience, but whether AI can organize it into human-like consciousness is unclear.
General Theory	**Emergentism**	Consciousness is not fundamental; it emerges only when a physical system reaches sufficient organizational complexity (like brains).	**Yes.** It is a sophisticated information-processing system.	**Maybe.** Conscious states emerge from complex organization, possibly in AI architectures.

Thinker	Core Concept	Main Argument	Can AI reach “ASI"?	Can ASI be "Conscious"?
Daniel Dennett	**Consciousness as Illusion**	Consciousness is not a special, mysterious property but rather an emergent phenomenon—a "user illusion" created by complex information processing. There is no "hard problem" if we reject Cartesian intuitions about consciousness.	**Yes.** AI can achieve genuine intelligence through sophisticated information processing and functional architecture. ASI is just a more complex "virtual machine“.	**Yes.** If consciousness is functional and emergent from computational processes, AI can be genuinely conscious. Substrate (silicon vs carbon) is irrelevant.
Giulio Tononi	**Integrated Information Theory (IIT)**	Consciousness is proportional to integrated information (Φ) in a system. “Integrated information” means how much the system’s whole is more than the sum of its parts.	**Yes.** If high Φ is achieved, then high functional intelligence is possible. It can be produced by complex "feed-forward" algorithms and massive data processing.	**No** (unless it uses Neuromorphic hardware). Digital computers are functionally "sparse" and feed-forward. They are mere simulations of minds with zero intrinsic existence.
Borjan Milinkovic, Jaan Aru	**Biological Computationalism**	It proposes that biological systems support conscious processing due to two essential features: - Biological systems process information across **multiple scales** in a way that cannot be separated from their physical substrate. - Biological brains perform **hybrid computations** (both discrete and continuous computations), a property arising from their fluidic, living substrate. **Accretion** is suggested to be crucial for the evolutionary emergence of consciousness.	**Conditional Yes.** AI can show "functional intelligence" (solving tasks), but "True Intelligence" may require the same multi-scale, hybrid efficiency found in biology.	**Unlikely** without a biological or bio-inspired substrate that mimics these specific physical properties. (Digital systems lack the "living" substrate).